

Design of a CMOS Bandgap Reference (BGR)

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1 Introduction

The objective of this project is to design a Bandgap Reference (BGR) based on the TSMC 180nm process. This report serves as a practical engineering log, documenting my design flow, reasoning, debugging processes, and key insights throughout the project. The design methodology references public textbooks and online materials.

2 Target Specifications

The goal is to design a highly stable BGR that meets the following constraints across all PVT corners and Monte Carlo (MC) mismatch variations:

Parameter	Target Specification
Temperature Coefficient (TC)	$< 20 \text{ ppm } (-40^\circ\text{C} \sim 125^\circ\text{C})$ (PVT)
PSRR @ DC	$< -65 \text{ dB}$ (PVT), $< -55 \text{ dB}$ (MC)
PSRR @ 100kHz	$< -50 \text{ dB}$ (PVT)
Phase Margin (PM)	$> 50^\circ$ (PVT)
RMS Noise (1kHz $\sim$ 100kHz)	$< 100 \mu\text{V}$ (PVT)
Output Offset (V_{OS})	$< 2\%$ @ 3σ (MC)
Current Consumption (I_{DC})	$< 300 \mu\text{A}$ (PVT)

3 Circuit Architecture & Analysis

The overall structure is shown in Figure 1. Instead of a traditional resistor, I utilize a PMOS current mirror to provide the biasing current. Assuming the operational amplifier (op-amp) maintains a virtual short at its input nodes, the voltage drop across resistor R_1 is driven by the base-emitter voltage difference:

$$V_{R1} = V_{BE1} - V_{BE2} = \Delta V_{BE} = V_T \ln(n) \quad (1)$$

This establishes a Proportional-To-Absolute-Temperature (PTAT) current $I_{PTAT} = V_T \ln(n)/R_1$. This current is mirrored by M3 to the output branch, generating the bandgap voltage:

$$V_{BG} = V_{BE3} + \left(\frac{R_2}{R_1}\right) V_T \ln(n) \quad (2)$$

Since V_{BE3} exhibits a negative temperature coefficient (CTAT) and V_T exhibits a positive temperature coefficient (PTAT), we can theoretically achieve a temperature-independent V_{BG} by tuning the multiplier term $(R_2/R_1) \ln(n)$.

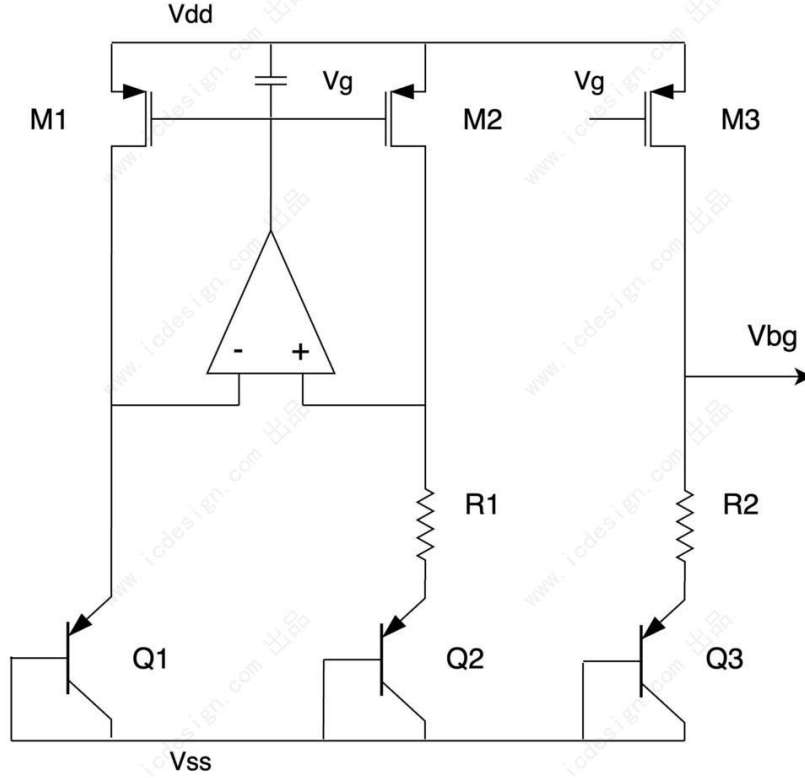


Figure 1: Core Architecture of the Bandgap Reference

4 Temperature Drift Optimization

Temperature drift is the most critical metric for a BGR. Our specification demands a $TC < 20$ ppm. The temperature coefficient is defined as:

$$TC = \frac{V_{BG,max} - V_{BG,min}}{V_{BG,avg} \cdot (T_{max} - T_{min})} \quad (3)$$

Given our temperature range (-40°C to 125°C) and target nominal voltage ($\approx 1.25\text{V}$), a 20 ppm drift corresponds to a maximum voltage variation of roughly 4 mV.

4.1 Single-Point vs. Global Compensation

The PTAT term $\partial \Delta V_{BE} / \partial T = (k/q) \ln(n)$ is nearly constant, meaning its second-order temperature effect is negligible. However, the CTAT term $\partial V_{BE} / \partial T$ is a non-linear function of temperature T . Consequently, perfect cancellation across the entire temperature range is impossible.

We have two primary compensation strategies:

- **Single-Point Compensation:** The PTAT and CTAT coefficients perfectly cancel at one specific temperature T_0 . On the drift curve, this corresponds to the point where the slope is zero.
- **Global Compensation (Chosen):** The net positive and negative voltage drifts are balanced over the entire specified temperature range (-40°C to 125°C). The drift curve will show equal output voltages at both temperature extremes. This method yields the smallest total variation across the full range.

I selected BJT emitter sizes of $10\mu\text{m} \times 10\mu\text{m}$ for all devices and set the ratio $n = 8$ for Q2. By running a DC temperature sweep and observing the total variation of ΔV_{BE} and V_{BE} across the full range, I calculated the ideal resistance ratio K :

$$K = \frac{R_2}{R_1} = \frac{\Delta(V_{BE})}{\Delta(\Delta V_{BE})} = 9.1953 \quad (4)$$

Simulating the circuit with ideal components and this K value yielded a total variation of $941\mu\text{V}$.

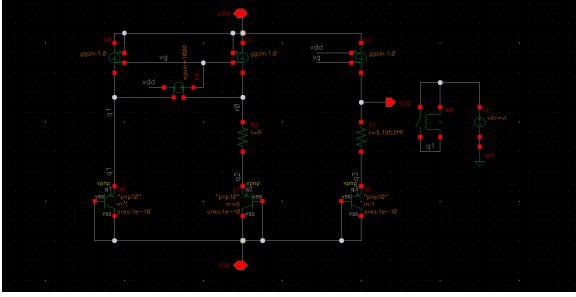


Figure 2: Ideal Schematic for K Calculation

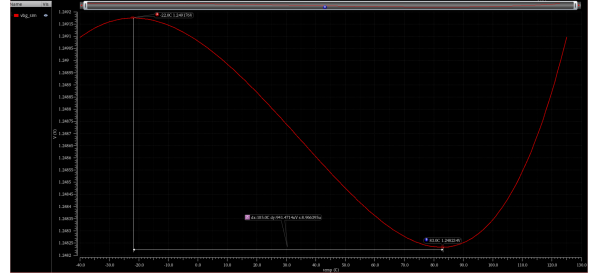


Figure 3: Simulated Voltage Drift Curve ($941\mu\text{V}$)

4.2 BJT Bias Point Selection

Minimizing the Second-Order Effect

The severity of V_{BE} 's second-order temperature non-linearity depends strongly on the BJT's operating point (i.e., its current density). By scanning the bias current $I_{PTAT} = \Delta V_{BE}/R_1$, we can locate a sweet spot that minimizes this curvature. However, whenever we change I_{PTAT} (by tuning R_1), the required global compensation ratio K also shifts. To automate this search, I configured a theoretical output node in Cadence: $V_{BG,cal}(T) = V_{BE}(T) + K \cdot \Delta V_{BE}(T)$, where K dynamically updates based on R_1 .

Scanning R_1 revealed that the theoretical output $V_{BG,cal}$ achieved its minimum temperature drift at $R_1 = 1.5\text{ k}\Omega$, which corresponded to a dynamically calculated ratio of $K = 9.54$. I then implemented these two specific values to construct and simulate the actual circuit, which yielded a practical V_{BG} temperature drift of $711\mu\text{V}$.

4.3 Implementing Real Resistors & Eliminating Channel-Length Modulation

When translating the design to physical layout components, I replaced the ideal R_1 and R_2 with process-specific resistor models. To minimize layout mismatch, R_2 is constructed using multiple R_1 unit cells in series and parallel.

Because the physical resistors possess a negative temperature coefficient (which exacerbates the PTAT current's positive drift), I slightly rounded K down to 9.5 to preemptively compensate. R_2 is thus formed by 9 series units and 2 parallel units of R_1 .

Furthermore, replacing the ideal VCCS at the top with a physical PMOS current mirror introduces Channel-Length Modulation. Since the drain of M2 sees a much larger resistance (R_2) than the drains of M0/M1 (R_1), the V_{DS} mismatch degrades current copying accuracy and ruins the temperature drift. To eliminate this, I intentionally inserted dummy resistors (matching R_2) below the drains of M0 and M1, perfectly balancing the voltage headroom and nullifying the CLM effect.

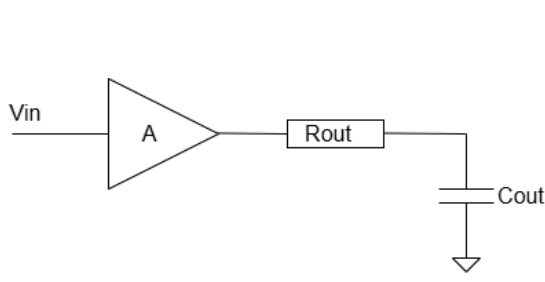


Figure 5: Simplified Op-amp Model for $A(s)$

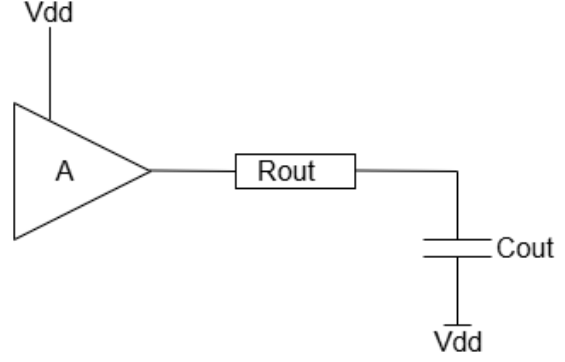


Figure 6: Simplified Op-amp Model for $b(s)$

Substituting $A(s)$ and $b(s)$ into the BGR PSRR equation yields:

$$\frac{V_{BG}}{V_{DD}}(s) = \left(\frac{K}{A}\right) \left(\frac{1}{G} + 1 - \beta + \frac{s}{G\omega_0}\right) \quad (8)$$

At high frequencies, the term $(1/G + 1 - \beta)$ becomes negligible compared to the frequency-dependent term. By equating the magnitude to 1, we derive the Unity-Gain Bandwidth of the PSRR:

$$GBW_{PSRR} = \frac{A \cdot G \cdot \omega_0}{K} \implies GBW_{PSRR} = \left(\frac{G}{K}\right) GBW_{opamp} \quad (9)$$

Rearranging the full transfer function into a standard pole-zero format:

$$\frac{V_{BG}}{V_{DD}}(s) = \left(\frac{K}{A}\right) \left(1 - \beta + \frac{1}{G}\right) \left[1 + \frac{s}{G\omega_0(1/G + 1 - \beta)}\right] \quad (10)$$

As we observe from the equations, we could theoretically tweak the op-amp so that β perfectly cancels $(1 + 1/G)$ to achieve infinite low-frequency PSRR. However, doing so pushes the zero frequency to near-DC, offering zero improvement to high-frequency PSRR. Therefore, to genuinely improve high-frequency PSRR, we must directly increase GBW_{PSRR} . This is achieved by either maximizing the op-amp's GBW or increasing the PMOS current mirror's intrinsic gain G .

6 Operational Amplifier Design

6.1 Topology Selection

Based on the derivation above, we require an op-amp topology where $\beta \approx 1$. Let's compare two common active-current-mirror OTAs (Figure 7).

- **NMOS Input Pair:** $PSRR = \frac{g_{m2}g_{m4}r_{o0}r_{o5}}{1 + g_{m2}g_{m4}r_{o0}r_{o5}} \approx 1$.
- **PMOS Input Pair:** M5 is biased by a constant current source, presenting an input impedance of r_{o5} . The PSRR is approximately $\frac{1}{2r_{o5}g_{m4}} \ll 1$.

Clearly, the NMOS input topology meets our requirement better. However, evaluating the required input common-mode range: $V_{in,cm} = V_{BE} \approx 0.7V \sim 0.8V$. For the standard NMOS OTA, $V_{in,cm} \geq V_{ov5} + V_{gs1}$. A 0.7V common-mode level is insufficient to keep the tail current source saturated.

To resolve this, I implemented a **Folded-Cascode** topology with an NMOS input pair. This architecture comfortably supports the low input common-mode voltage while simultaneously boosting the DC gain, heavily benefiting the loop dynamics.

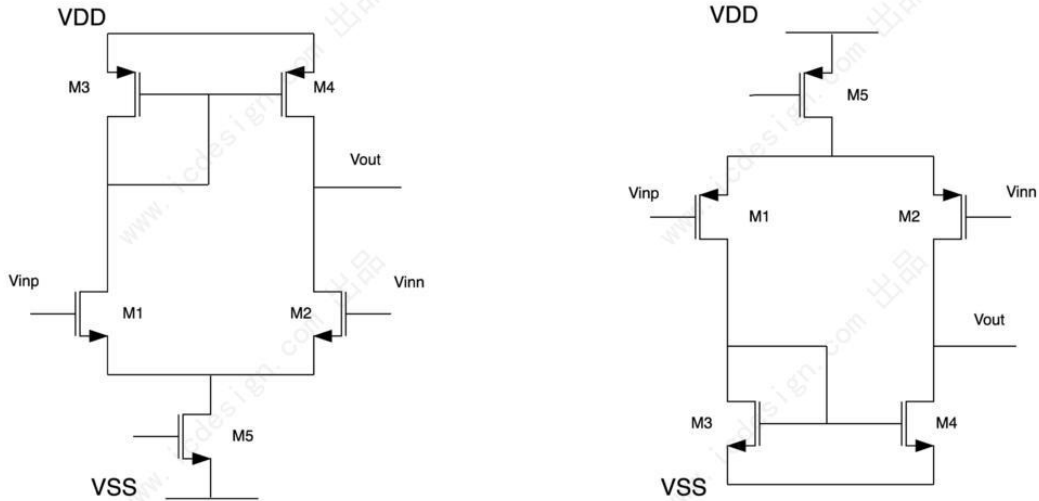


Figure 7: NMOS vs PMOS Input OTAs

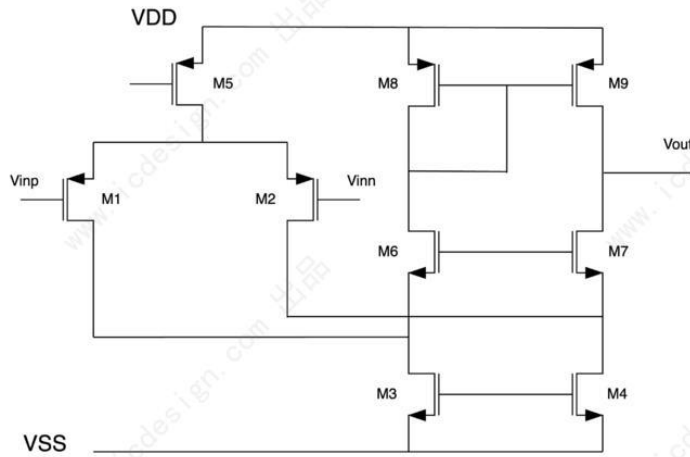


Figure 8: Selected Folded-Cascode Topology

6.2 Op-amp Design & Biasing

Based on simulation using a VCVS to represent the op-amp (Figure ??), I extracted the target operating points: the input common-mode voltage is roughly 728 mV, and the required output common-mode voltage is roughly 1.2V. I sized the folded-cascode op-amp accordingly to match these quiescent points before closing the loop.

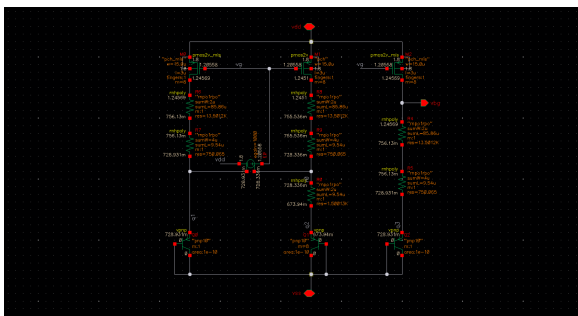


Figure 9: Operating Point Extraction

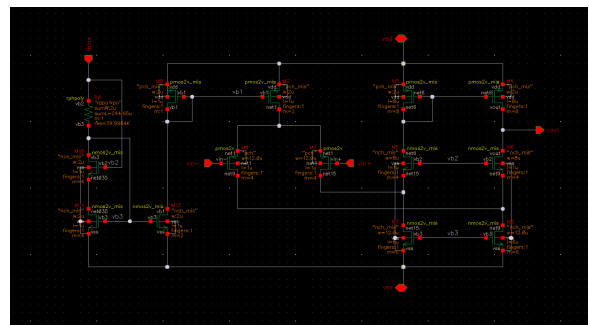


Figure 10: Folded-Cascode Schematic

6.3 Small-Signal PSRR Derivation of the Folded-Cascode

To understand why op-amp offset degrades the BGR's PSRR and why low V_{DD} corners fail, we rigorously derive the folded-cascode's PSRR (β). The folded-cascode has two primary paths from V_{DD} : the tail current source (M5) and the current mirror loads (M8, M9). We apply superposition to find the total V_{out}/V_{DD} .

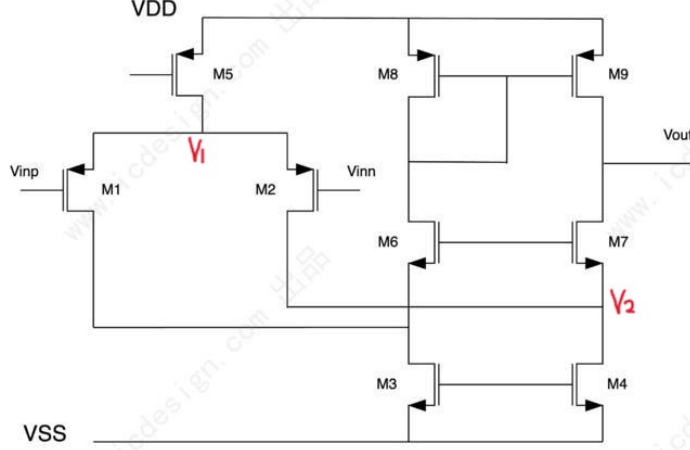


Figure 11: Op-amp Schematic for PSRR Analysis (Annotations match Section 6.3)

6.3.1 Tail Current Source Feedthrough

$$\frac{V_1}{V_{DD}} = \frac{1/(2g_{m1})}{r_{o5} + 1/(2g_{m1})} = \frac{1}{1 + 2g_{m1}r_{o5}} \quad (11)$$

$$\Delta I_1 = \frac{V_{DD} - V_1}{r_{o5}} = \frac{V_{DD} \cdot 2g_{m1}}{1 + 2g_{m1}r_{o5}} \quad (12)$$

$$\Delta V_2 = 0.5 \cdot \Delta I_1 \cdot \left(r_{o4} \parallel \frac{1}{g_{m7}} \right) = \frac{0.5 \cdot \Delta I_1 \cdot r_{o4}}{1 + g_{m7}r_{o4}} \quad (13)$$

The current injected into the top mirror is $\Delta I_2 = \Delta V_2 \cdot g_{m7}$. Since this is a common-mode fluctuation, the drain of M9 tracks M8 (virtual short), making M9's input impedance $1/g_{m9}$.

$$\Delta V_{out} = \frac{\Delta I_2}{g_{m9}} \quad (14)$$

$$\left(\frac{V_{out}}{V_{DD}} \right)_{tail} = \left[\frac{g_{m1}}{1 + 2g_{m1}r_{o5}} \right] \left[\frac{g_{m7}r_{o4}}{1 + g_{m7}r_{o4}} \right] \frac{1}{g_{m9}} \approx \frac{1}{2g_{m9}r_{o5}} \approx 0 \quad (15)$$

6.3.2 Current Mirror Feedthrough

Under common-mode stimulus, M5 splits into two parallel devices, halving the current and doubling r_{o5} . The impedance looking down from the output is $R_{down} = g_{m7}r_{o7}(r_{o4} \parallel 2g_{m2}r_{o2}r_{o5})$. The impedance looking up remains $R_{up} = 1/g_{m9}$.

$$\left(\frac{V_{out}}{V_{DD}} \right)_{mirror} = \frac{R_{down}}{R_{up} + R_{down}} \approx 1 \quad (16)$$

Summing both components, the overall op-amp PSRR is $\beta \approx 1$, validating our topology choice.

6.4 Analysis of Op-amp Offset Degradation

Offset severely degrades BGR performance. We analyze its impact on two fronts.

6.4.1 Impact on PSRR

If the op-amp's natural output common-mode voltage deviates from the ideal bias point of the PMOS gates ($V_{out,cm} \neq V_{g8,cm}$), it introduces an equivalent input offset: $V_{os} = (V_{out,cm} - V_{g8,cm})/A$. This offset breaks the symmetry: $V_{GS1} \neq V_{GS2} \Rightarrow g_{m1} \neq g_{m2}$. When a common-mode ripple arrives from V_{DD} via M5, it is no longer split equally. This mismatch generates a differential current fluctuation ΔI_{diff} at the output stage, which is subsequently multiplied by the high output resistance r_{o9} .

$$\left(\frac{V_{out}}{V_{DD}} \right)_{dm} = r_{o9} \cdot (\Delta I_1 - \Delta I_2) \quad (17)$$

This differential fluctuation pulls β far away from 1, destroying the BGR's PSRR.

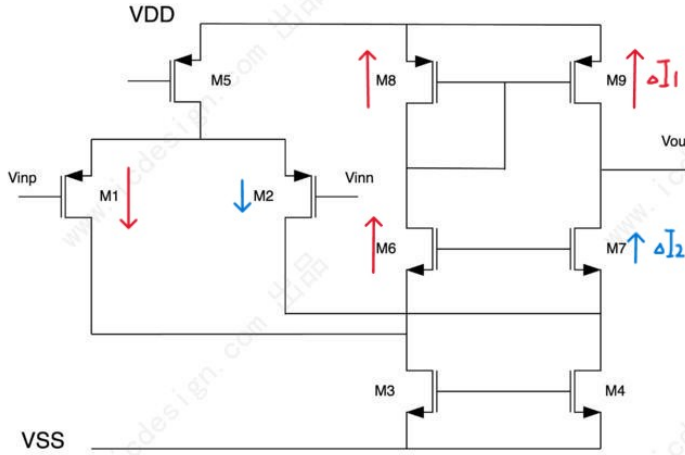


Figure 12: Offset-Induced Differential Fluctuation in PSRR

6.4.2 Impact on Output Voltage & TC

Assuming the op-amp maintains a virtual short, $V_B = V_A = V_{BE1}$, and $V_C = V_{BE1} + V_{os}$. The voltage across R_1 becomes:

$$V_{R1} = V_{BE1} + V_{os} - V_{BE2} = \Delta V_{BE} + V_{os} \quad (18)$$

The final BGR output evaluates to:

$$V_{BG} = V_{BE} + K(\Delta V_{BE} + V_{os}) \quad (19)$$

The op-amp offset voltage is amplified by K (approx. 9.5) and directly superimposes on the output voltage.

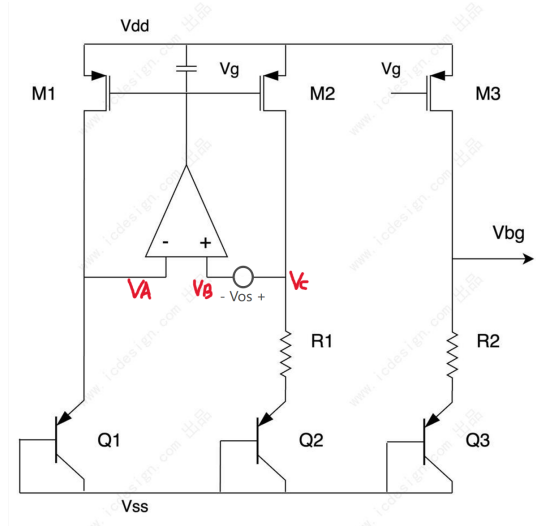


Figure 13: Injection of Op-amp Offset (V_{os}) into the BGR Core

- **Systematic Offset:** Because the op-amp controls the PTAT current via the PMOS gates, the op-amp's output voltage is inherently CTAT. Since $V_{os} = (V_{out,cm} - V_{cm,ideal})/A$, the systematic V_{os} is also CTAT and varies with temperature. When this temperature-dependent voltage is superimposed on the output, it severely degrades the temperature drift.
- **Random Offset:** Thanks to the BGR's high PSRR and low temperature drift, the output voltage is nearly immune to PVT variations, making the absolute output precision dominated by mismatch. Random offset is temperature-independent (acting as a pure DC offset), causing the absolute output voltage to vary across different chips.

Canceling the Temperature Coefficient of Systematic Offset

Since the systematic V_{os} fluctuates with temperature and degrades the TC, can we eliminate its temperature coefficient? We could bias the op-amp using the BGR's self-generated PTAT current. Suppose the temperature rises: I_{PTAT} increases, which requires the op-amp's output voltage to drop. If a constant-current bias is used, the input offset voltage must change to drive this output shift. However, if a PTAT bias is used, as the op-amp's bias current (I_{PTAT}) increases, its output voltage naturally drops. This requires no change at the input, effectively eliminating the temperature coefficient of the input offset voltage and improving the overall TC. While mathematically elegant, I opted not to adopt this method in this design.

Therefore, we must address the offsets deterministically:

- Eliminating Systematic Offset:** I used parameter sweeps to precisely size M8 and M9, ensuring $V_{DS9} = V_{GS9}$ within the closed loop to nullify the systematic offset.
- Minimizing Random Offset:** I increased the channel width (W) of the input transistors and the channel length (L) of the load transistors to suppress random mismatch.

7 Noise Optimization

The noise contributors in the circuit can be categorized into the op-amp's equivalent input noise, the PMOS current mirror noise, the resistor noise, and the BJT noise. Assuming a virtual short

at the op-amp inputs, these noise sources are transferred to the output with the same transfer functions as the offset voltage analyzed earlier:

$$V_{n,BG} = K \cdot V_{n,op} \quad (\text{Op-amp equivalent input noise}) \quad (20)$$

$$V_{n,BG} = K \cdot V_{n,R1} + V_{n,R2} \quad (\text{Resistor noise}) \quad (21)$$

$$V_{n,BG} = K \cdot I_{n,M1} \cdot \left(\frac{1}{g_{m1}} \right) \quad (\text{PMOS M1 noise}) \quad (22)$$

$$V_{n,BG} = K \cdot I_{n,M2} \cdot \left(R_1 + \frac{1}{g_{m2}} \right) \quad (\text{PMOS M2 noise}) \quad (23)$$

$$V_{n,BG} = I_{n,M3} \cdot \left(R_2 + \frac{1}{g_{m3}} \right) \quad (\text{PMOS M3 noise}) \quad (24)$$

(BJT noise is entirely negligible due to their massive transconductance).

Simulation confirms that the op-amp dominates the overall noise profile. Specifically, the major contributors are the thermal noise from the input transistors and the NMOS bias current sources. We can suppress this noise by increasing the transconductance (g_m) of the input transistors and decreasing the g_m of the current sources.

Results Display Window			
Window Expressions Info Help			
Device	Param	Noise Contribution	% Of Total
I6.I15.M4.m0	id	1.79014e-05	13.15
I6.I15.M3.m0	id	1.78733e-05	13.11
I6.I15.M10.m0	id	1.55862e-05	9.97
I6.I15.M9.m0	id	1.55856e-05	9.97
I6.R0.Rpure	thermal_noise	1.54855e-05	9.84
I6.M1.m0	id	1.44238e-05	8.54
I6.I15.M12.m0	id	1.44064e-05	8.52
I6.I15.M11.m0	id	1.43849e-05	8.49
I6.M2.m0	id	9.66815e-06	3.84
/I6/Q0	ic	6.7868e-06	1.89
/I6/Q1	ic	6.64919e-06	1.81
I6.I15.M4.m0	fn	6.23035e-06	1.59
I6.I15.M3.m0	fn	6.22134e-06	1.59
I6.M0.m0	id	4.77243e-06	0.93
I6.M1.m0	fn	4.74146e-06	0.92
I6.I15.M10.m0	fn	4.51184e-06	0.84
I6.I15.M9.m0	fn	4.51179e-06	0.84
/I6/Q1	ib	4.07706e-06	0.68
/I6/Q0	ib	4.00979e-06	0.66
I6.M2.m0	fn	3.1783e-06	0.41
I6.R0.Rend1	thermal_noise	2.49775e-06	0.26
I6.R0.Rend2	thermal_noise	2.49775e-06	0.26
/I6/Q0	rb	2.44969e-06	0.25
I6.I15.M12.m0	fn	2.40126e-06	0.24
I6.I15.M11.m0	fn	2.39781e-06	0.24
I6.M0.m0	fn	1.56882e-06	0.10
I6.R49_8_dmy0.Rpure	thermal_noise	1.50997e-06	0.09
I6.R49_6_dmy0.Rpure	thermal_noise	1.50997e-06	0.09
I6.R49_4_dmy0.Rpure	thermal_noise	1.50997e-06	0.09
I6.R49_7_dmy0.Rpure	thermal_noise	1.50997e-06	0.09
Integrated Noise Summary (in V) Sorted By Noise Contributors			
Total Summarized Noise = 4.93594e-05			
No input referred noise available			
The above noise summary info is for noise data			

Figure 14: Noise Contribution Summary

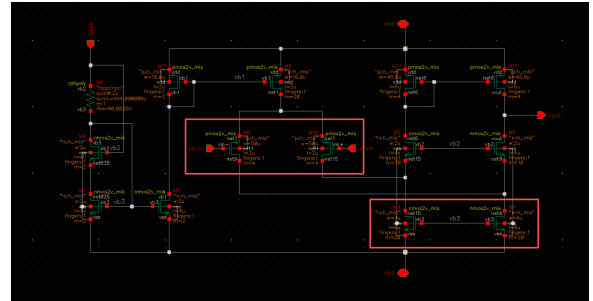


Figure 15: Op-amp Noise Contributors (Highlighted)

8 Loop Stability & Compensation

The BGR's dominant pole lies at the op-amp's output, and the non-dominant pole is at the op-amp's input nodes. Standard dominant-pole compensation (adding C_C to ground) would lower

the loop's GBW to achieve Phase Margin. However, as proven in Section 5.2, $GBW_{PSRR} \propto GBW_{opamp}$. Throttling the op-amp's bandwidth directly kills high-frequency PSRR.

To bypass this tradeoff, we can introduce a nulling resistor R_z in series with C_C . The transfer function becomes (Figure 17):

$$H(s) = \frac{1 + sR_zC_C}{1 + sR_zC_C + sC_{out}R_{out} + s^2C_{out}C_C R_{out}R_z} \quad (25)$$

This injects a Left-Half-Plane (LHP) zero at $\omega_z = 1/(R_zC_C)$, which provides phase lead without crushing the bandwidth. However, running a baseline STB simulation revealed that the uncompensated circuit already boasts a robust Phase Margin of 58° . Thus, explicit RC compensation was deemed unnecessary and omitted to save area.

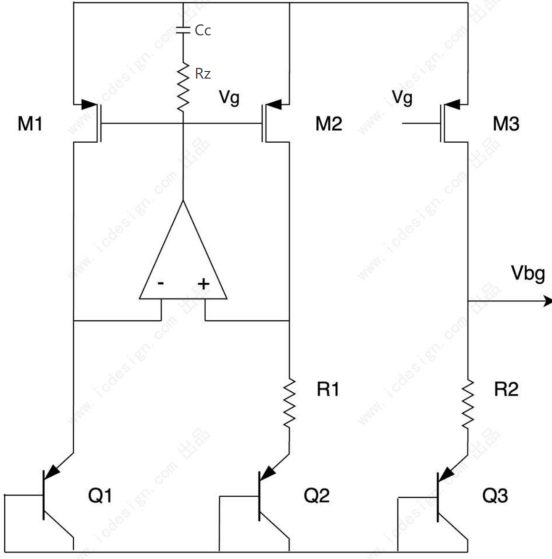


Figure 16: Compensation Topology

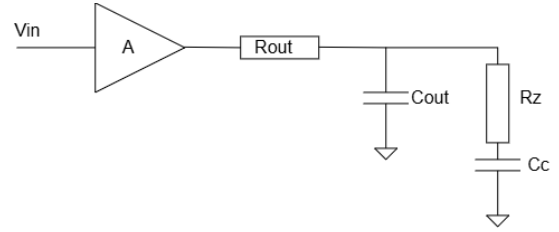


Figure 17: Small-Signal Model

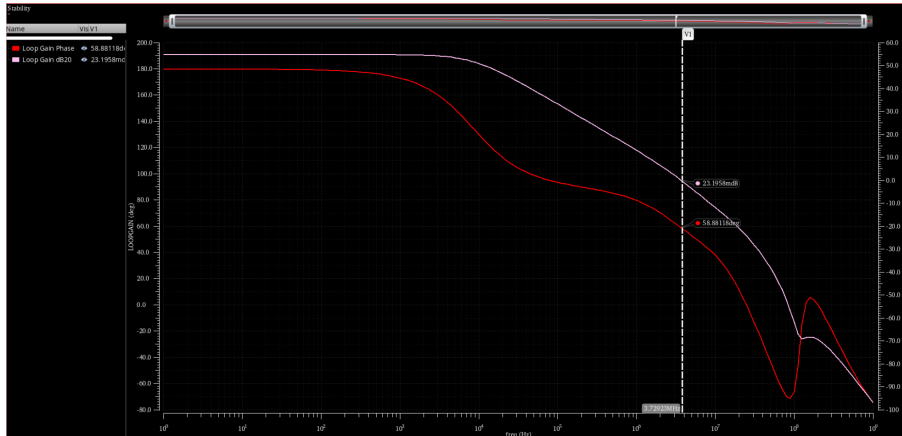


Figure 18: STB Bode Plot (PM = 58°)

9 Simulation & Iterative Optimization

9.1 Monte Carlo Simulation

The MC constraints required low-frequency PSRR < -55 dB and $V_{OS} < 2\%$ @ 3σ (meaning the standard deviation of V_{BG} must be < 8 mV). By methodically up-sizing the W and L of critical

mismatch contributors (input and load pairs), the design comfortably passed MC requirements.

Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 99.5 % (199 passed/200 pts) Confidence Level: <not set> Filter: <not set>									
-	bandgap:ch10_bg_sim_mc:1								
-	PSRR_dc(summary)	99.5	-117.6		-65.47	-80.85	9.131	0.944	0
	PSRR_dc	99.5	-117.6	< -55	-65.47	-80.85	9.131	0.944	0
-	vbg(summary)	99.5	1.242		1.265	1.254	5.185m		0
	vbg	99.5	1.242	info	1.265	1.254	5.185m		0

Figure 19: Monte Carlo Yield and Variance Results

9.2 PVT Variations & Debugging

I subjected the design to a 27-corner PVT sweep: $V_{DD} \in \{1.6V, 1.8V, 2.0V\}$, $Temp \in \{-40^\circ C, 27^\circ C, 125^\circ C\}$, and Process $\in \{tt, ff, ss\}$.

Debugging PSRR Degradation at Low VDD

During the PVT sweep, I observed a severe PSRR degradation at several $V_{DD} = 1.6V$ corners. I suspected that the low supply voltage was forcing certain transistors into the linear (triode) region. Checking the DC operating points confirmed this hypothesis: the PMOS current mirror at the top of the BGR and the tail current source above the op-amp's input pair were on the verge of entering the linear region due to reduced voltage headroom. From the derivation in Section 5.1, we know the BGR's PSRR is heavily dependent on the PMOS mirror's intrinsic gain (G). Meanwhile, Section 6.3.1 shows that the op-amp's PSRR (β) is inversely proportional to the tail source's output resistance (r_{o5}). As these transistors approach the linear region, their output impedances (r_o) plummet, causing both $1/G$ and β to spike, which drastically degrades the overall BGR PSRR.

The Fix: I increased the W/L ratios of these specific PMOS transistors to reduce their overdrive voltages and secure their saturation margins at low V_{DD} . Because this sizing change altered the loop's biasing, I subsequently re-scanned the resistor ratio K to ensure global temperature compensation was maintained, and simultaneously re-tuned the W/L of the op-amp's active current mirror (M8/M9) to minimize the systematic offset under the new conditions.

After these structural adjustments, the circuit successfully passed all 27 PVT corners.

Output	History	Test	Conditions	Min	Max	Std Dev	Spec	Pass/Fail
delta_vbg	Interactive.34	bandgap:ch10_bg_sim_mc:1	vdd=1.6,1.8,2 temperature=-40,27,125 CornerModels=ff_res,ss_res,tt_res	635.4u	2.682m	923.6u	< 4.125m	pass
PSRR_dc	Interactive.34	bandgap:ch10_bg_sim_mc:1	vdd=1.6,1.8,2 temperature=-40,27,125 CornerModels=ff_res,ss_res,tt_res	-109.9	-69.4	9.959	< -65	pass
PSRR_100k	Interactive.34	bandgap:ch10_bg_sim_mc:1	vdd=1.6,1.8,2 temperature=-40,27,125 CornerModels=ff_res,ss_res,tt_res	-82	-50.43	5.847	< -50	pass
noise	Interactive.34	bandgap:ch10_bg_sim_mc:1	vdd=1.6,1.8,2 temperature=-40,27,125 CornerModels=ff_res,ss_res,tt_res	40.48u	60.68u	6.777u	< 100u	pass
Phase Margin	Interactive.34	bandgap:ch10_bg_sim_mc:1	vdd=1.6,1.8,2 temperature=-40,27,125 CornerModels=ff_res,ss_res,tt_res	55.39	62.4	1.899	> 50	pass
idc	Interactive.34	bandgap:ch10_bg_sim_mc:1	vdd=1.6,1.8,2 temperature=-40,27,125 CornerModels=ff_res,ss_res,tt_res	199.9u	287u	26.95u	< 300u	pass

Figure 20: Final 27-Corner PVT Sweep Summary Table

9.3 Final Circuit Implementation & Performance

The final optimized schematics for the BGR core and Op-amp are presented below. Performance at nominal conditions (1.8V, 27°C, TT) is summarized as follows:

- **Tempco (TC):** 635.6 μV
- **PSRR @ DC:** -94.5 dB
- **PSRR @ 100kHz:** -58.3 dB
- **Phase Margin:** 58.8°
- **RMS Noise:** 49.4 μV
- **DC Current:** 229 μA

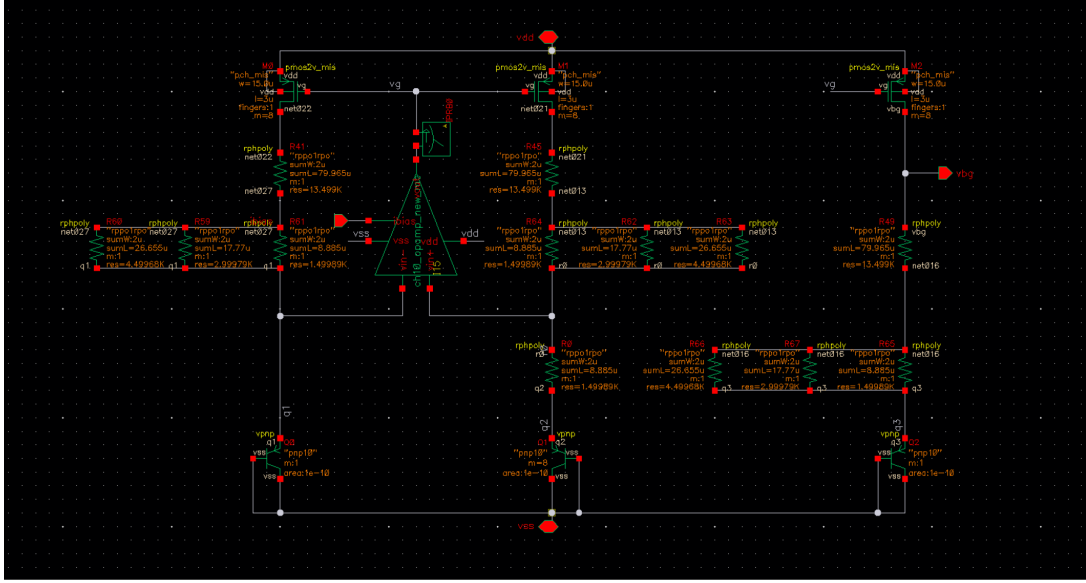


Figure 21: Final Complete BGR Core Schematic

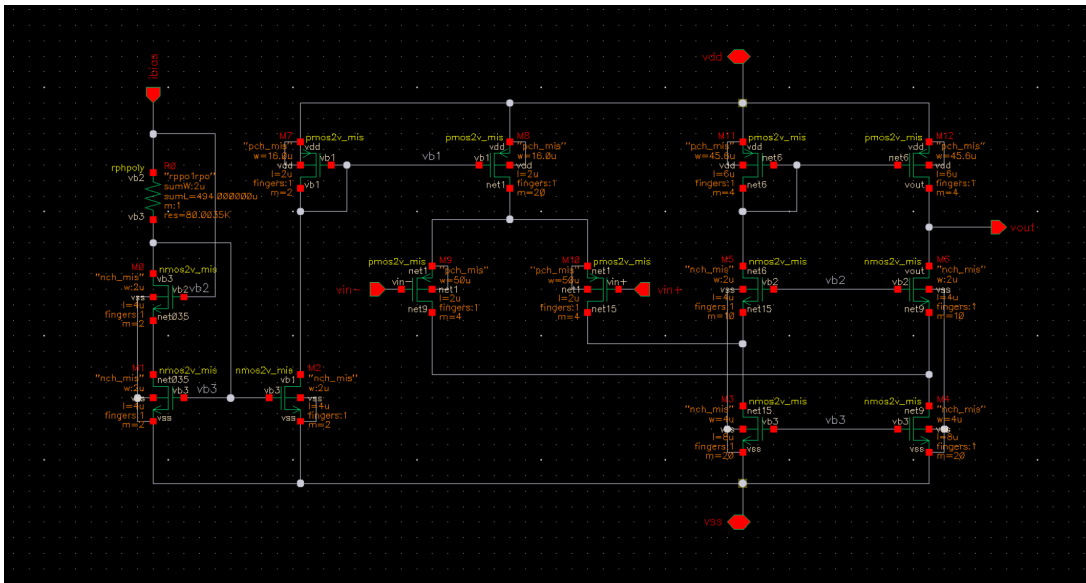


Figure 22: Final Complete Folded-Cascode Op-amp Schematic